

R Workshop

From Model to Meaning¹

Completing the Analytic Workflow: Why Fitting the Model is Just the Beginning

Dr. Daniel Lüdecke
d.luedecke@uke.de
<https://github.com/easystats>



Universitätsklinikum
Hamburg-Eppendorf

¹ Arel-Bundock, V. (2026). Model to meaning: How to interpret statistical models with R and Python. CRC Press, Taylor & Francis Group.
<https://doi.org/10.1201/9781003560333> (also free at <https://marginaleffects.com/>)

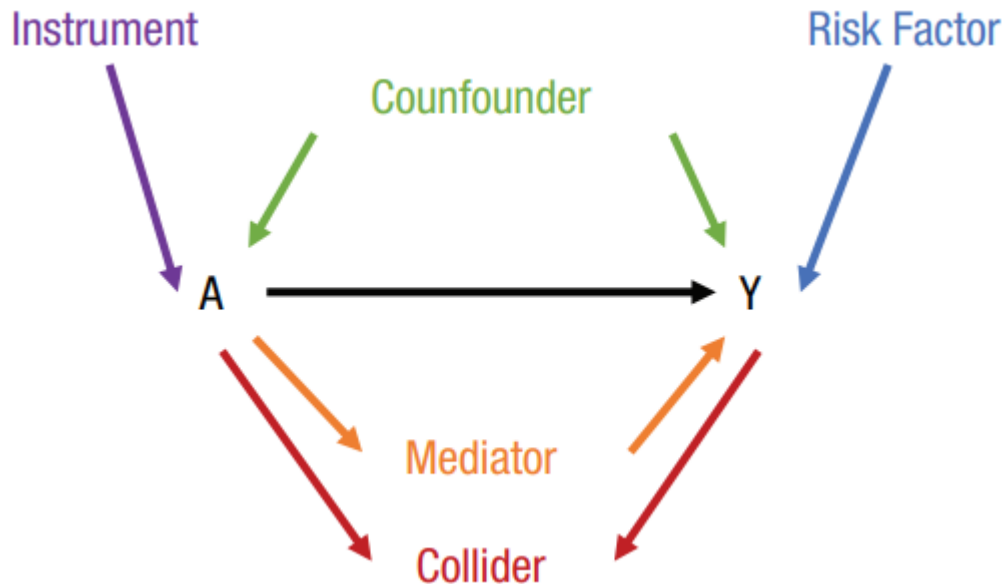
Part 1

Step 1: Setup the Model

- 1 What is your research question?**
Clearly define your **theoretical estimand** and assess if the sample matches your target.
- 2 Which predictors to include?**
Use **DAGs**: Include confounders & risk factors. Strictly exclude mediators & colliders to avoid bias.
- 3 How to interpret the output?**
Interpret coefficients as descriptive comparisons and **conditional effects**, not causal effects.

Which predictors to include?

- DAGs visually map out the theoretical causal assumptions behind a model.
- They serve as a practical guide to determine which variables should be included or excluded from the analysis.



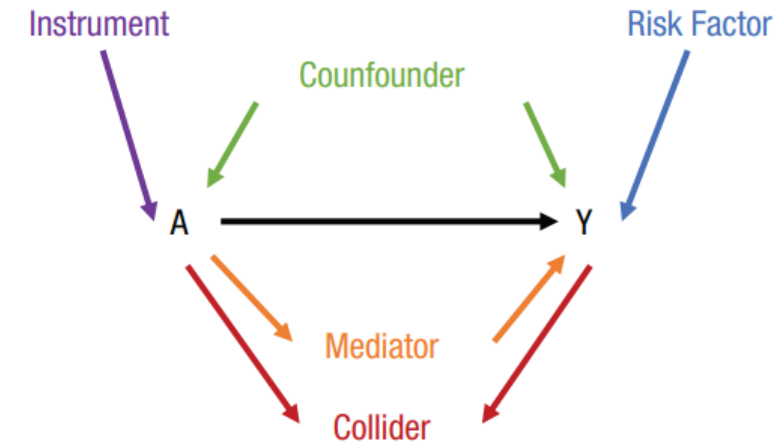
Which predictors to include?

Must include (to prevent bias)

- **Confounders:** Common causes of both the exposure and the outcome. You must control for them to close “backdoor paths” and avoid spurious correlations.

May include (to increase precision)

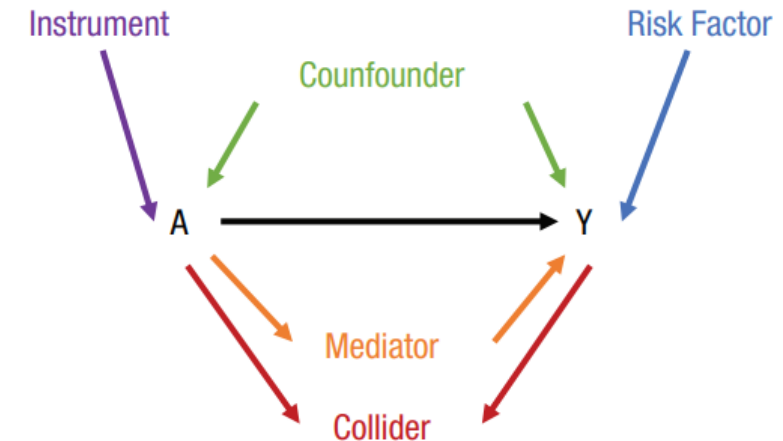
- **Risk Factors:** Variables that only affect the outcome. Controlling for them is not required to prevent bias, but it reduces error variance and increases the precision of your estimate.



Which predictors to include?

Must not include (overadjustment bias)

- **Mediators:** Variables on the direct “frontdoor path” from exposure to outcome. Controlling for them erroneously extracts the true causal effect.
- **Colliders:** Common consequences of both the exposure and the outcome. Controlling for a collider creates artificial distortion (collider bias).
- **Instrumental Variables:** Variables that only affect the exposure. Controlling for them massively amplifies the bias of any remaining unmeasured confounders (bias amplification).



Interpret coefficients as comparisons, not effects



Example

```

1 data(efc, package = "modelbased")
2 efc <- datawizard::to_factor(efc, c("e42dep", "c161sex"))
3 model <- lm(neg_c_7 ~ e42dep + c161sex + c160age, data = efc)
4 parameters::model_parameters(model, include_info = TRUE)
5 #> Parameter | Coefficient | SE | 95% CI | t(883) | p
6 #> -----
7 #> (Intercept) | 8.17 | 0.67 | [ 6.86, 9.49] | 12.17 | < .001
8 #> e42dep [slightly dependent] | 1.30 | 0.50 | [ 0.31, 2.28] | 2.59 | 0.010
9 #> e42dep [moderately dependent] | 2.69 | 0.49 | [ 1.74, 3.65] | 5.53 | < .001
10 #> e42dep [severely dependent] | 4.40 | 0.49 | [ 3.43, 5.37] | 8.92 | < .001
11 #> c161sex [Female] | 0.41 | 0.28 | [-0.14, 0.97] | 1.46 | 0.146
12 #> c160age | 0.01 | 9.30e-03 | [-0.01, 0.03] | 1.33 | 0.184
13 #>
14 #> Model: neg_c_7 ~ e42dep + c161sex + c160age (889 Observations)
15 #> Sigma: 3.581 (df = 883)
16 #> RMSE : 3.569
17 #> R2: 0.147; adjusted R2: 0.142
18 #>
19 #> Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
20 #> using a Wald t-distribution approximation.

```

Interpret coefficients in terms of *comparisons*

```
#> Parameter | Coefficient |
#> -----|-----|
#> (Intercept) | 8.17 |
#> e42dep [slightly dependent] | 1.30 |
#> e42dep [moderately dependent] | 2.69 |
#> e42dep [severely dependent] | 4.40 |
#> c161sex [Female] | 0.41 |
#> c160age | 0.01 |
```

- Estimates of categorical variables are by default interpreted in comparison to their **reference category**.
- In case of **e42dep**, the reference category is **independent**.
- The highlighted coefficient for **severely dependent** says: The *subjective burden of care* (**outcome**) is - on average - 4.4 points higher (**estimate**) for family carers who care for elderly that are *severely dependent* (**parameter**) compared to those carers who care for elderly who are *independent* (**reference category**).
(for family carers of *same* sex and age)



Interpret coefficients as comparisons, not effects

$$\text{Weight (tons)} = 2.4 + \underline{0.3}(\text{height}) + \dots$$



if all other variables constant, we expect a 1 foot taller dragon to weigh 0.3 tons more, on average.

@allisonhorst

Interpret coefficients in terms of *comparisons*

```
#> Parameter | Coefficient |
#> -----|-----|
#> (Intercept) | 8.17 |
#> e42dep [slightly dependent] | 1.30 |
#> e42dep [moderately dependent] | 2.69 |
#> e42dep [severely dependent] | 4.40 |
#> c161sex [Female] | 0.41 |
#> c160age | 0.01 |
```

- Estimates of *continuous* variables are by default interpreted in comparison to a *one-unit change*.
- The highlighted coefficient for **c160age** says: The *subjective burden of care* (**outcome**) increases - on average - by 0.01 points (**estimate**) when the *age of family carers* (**parameter**) increases by 1 year (unit of age is “years”).

(for family carers of *same* age and dependency level)



But does this answer my research question?

How to ask your model the right questions

Part 2

Step 2: Post-Estimation with *modelbased*

1 Predictor of Interest

Functions: `estimate_means()`, `estimate_slopes()`, `estimate_contrasts()`

2 Evaluation Points

Control exact levels via arguments: `by`, `contrast`, `slope`

3 Target Population ("Estimand")

Define generalizations via `estimate`:

Conditional: "specific", "typical"

Marginal: "average", "population"

One model, endless insights

- **The Goal:** Extract multiple insights from one comprehensive model
(From *exploratory* analysis to formal hypothesis *testing*)
- **The Prerequisite:** Build a model complex enough to capture your core theoretical assumptions.
- **The Tool:** Leveraging the R *modelbased* package for interpretation and post-estimation.

Does perceived stigmatization predict an increase in somatic symptom severity (PHQ-15)?

- Model Type: Linear mixed model for hierarchical, longitudinal data
- Dependent Variable: phq15 (Somatic symptom burden)
- Fixed Effects:
 - The interaction *stigma_unreal* * *time* tests whether the trajectory of PHQ-15 (time trend) differs between the various stigma groups
 - Controls for sex, z-standardized age, education, and migration history
- Random Effects:
 - 3-level nesting. Measurement time points are nested within patients, who are in turn nested within disease groups
 - Random Intercepts: Accounts for baseline differences; allows each patient and each disease group to have their own starting level of PHQ-15
 - Random Slopes (time): Accounts for individual temporal developments; allows the time trend (slope) to vary by patient and disease group

The `modelbased` Toolkit: Core Functions & Marginalization

Core Functions of the Package

- **`estimate_means()`**: Calculates the model-based average response (marginal means) at specific predictor values or factor levels (e.g., the expected PHQ-15 score for a specific stigma group).
- **`estimate_slopes()`**: Estimates the marginal effects or trends of continuous predictors, allowing you to assess how the effect of time changes depending on other conditions (e.g., the slope of symptoms over time).
- **`estimate_contrasts()`**: Calculates and tests the differences (contrasts) between those marginal means or slopes, which is essential for formal hypothesis testing (e.g., pairwise comparisons between groups).

The `modelbased` Toolkit: Core Functions & Marginalization

Why use `estimate="average"`?

- By default, models often estimate marginal means by holding all other covariates (like age or sex) constant at an artificial “average” value (a reference grid).
- Setting `estimate="average"` changes the workflow. The model first predicts the expected outcome for *every single actual observation* in your dataset based on their real covariate values.
- These individual predictions are then averaged by the groups of interest. This ensures that the resulting estimates accurately reflect the true, underlying distribution and demographic composition of the actual sample at hand, rather than a hypothetical average.
- `estimate="population"` is another option.

Does perceived stigmatization predict an increase in somatic symptom severity (PHQ-15)?

- R code for data preparation:

```
library(easystats)

# load data
stigma <- data_read("stigma_reduced.rds")

# some recoding: for personal reasons, "time" should start with 0,
# and age should be standardized
stigma$time <- slide(stigma$time3)
stigma <- standardize(stigma, "age", append = TRUE)

# we recode the disagreement and agreement levels into one category
stigma$stigma_unreal <- recode_into(
  stigma_1_unreal %in% c("strongly disagree", "disagree") ~ "(strongly) disagree",
  stigma_1_unreal %in% c("strongly agree", "agree") ~ "(strongly) agree",
  data = stigma
)
```

Does perceived stigmatization predict an increase in somatic symptom severity (PHQ-15)?

- R code for mixed model:

```
library(glmmTMB)

m1 <- glmmTMB(
  # interaction term to model trajectories of phq15 by stigma groups
  phq15 ~ stigma_unreal * time +
  # controls
  sf_sex +
  age_z +
  education_casmin +
  migration_history +
  # higher level groups: patients (repeated measures) nested in (i.e.
  # classified into) disease groups. we assume varying time slopes for
  # both groups, i.e. trend of phq15 may vary between patients
  # and disease groups
  (1 + time | disease_group / patid),
  data = stigma
)
```



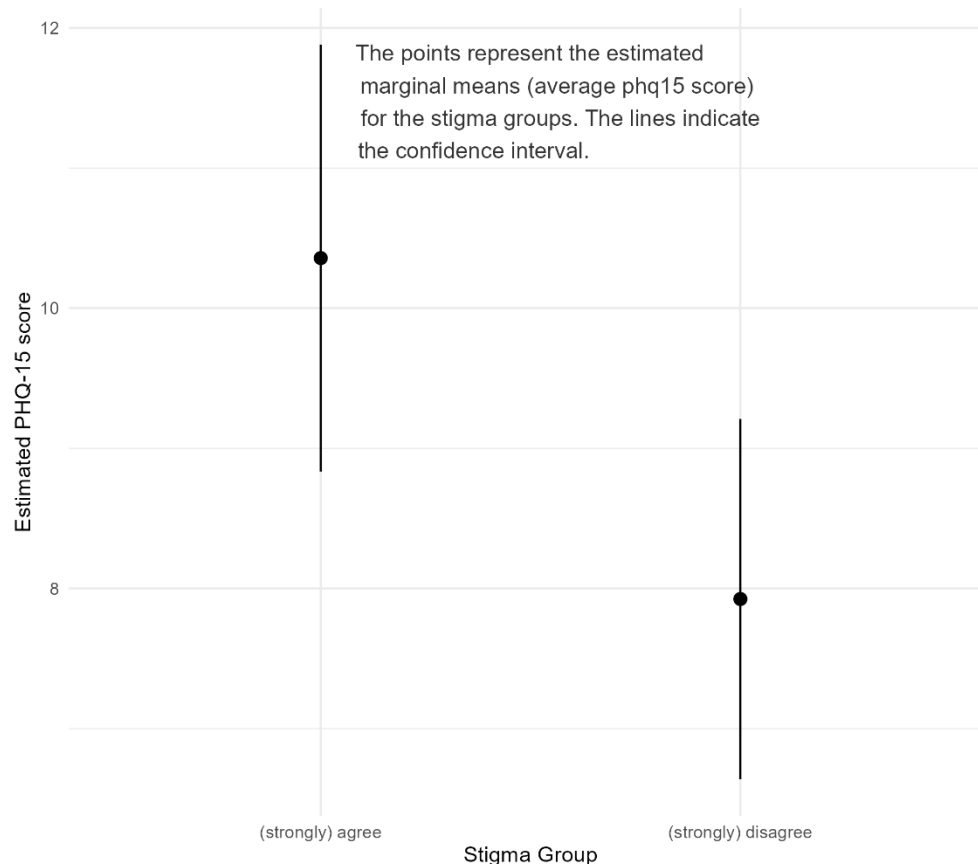
Predicted overall PHQ-15 scores: Comparing stigma groups

How does symptom severity differ between stigma groups?

Estimated Marginal Means

Run: `estimate_means(m1, by = 'stigma_unreal')`

This command calculates the estimated marginal means for each stigma group, comparing their overall average score.



```
1 estimate_means(
2   m1,
3   by = "stigma_unreal",
4   estimate = "average"
5 )
6 #> Average Predictions
7 #>
8 #> stigma_unreal      | Mean | SE | 95% CI | z
9 #> -----
10 #> (strongly) agree   | 10.36 | 0.78 | [8.83, 11.88] | 13.33
11 #> (strongly) disagree | 7.93 | 0.66 | [6.64, 9.21] | 12.09
12 #>
13 #> Variable predicted: phq15
14 #> Predictors modulated: stigma_unreal
```

Predicted PHQ-15 trajectories: Comparing stigma groups

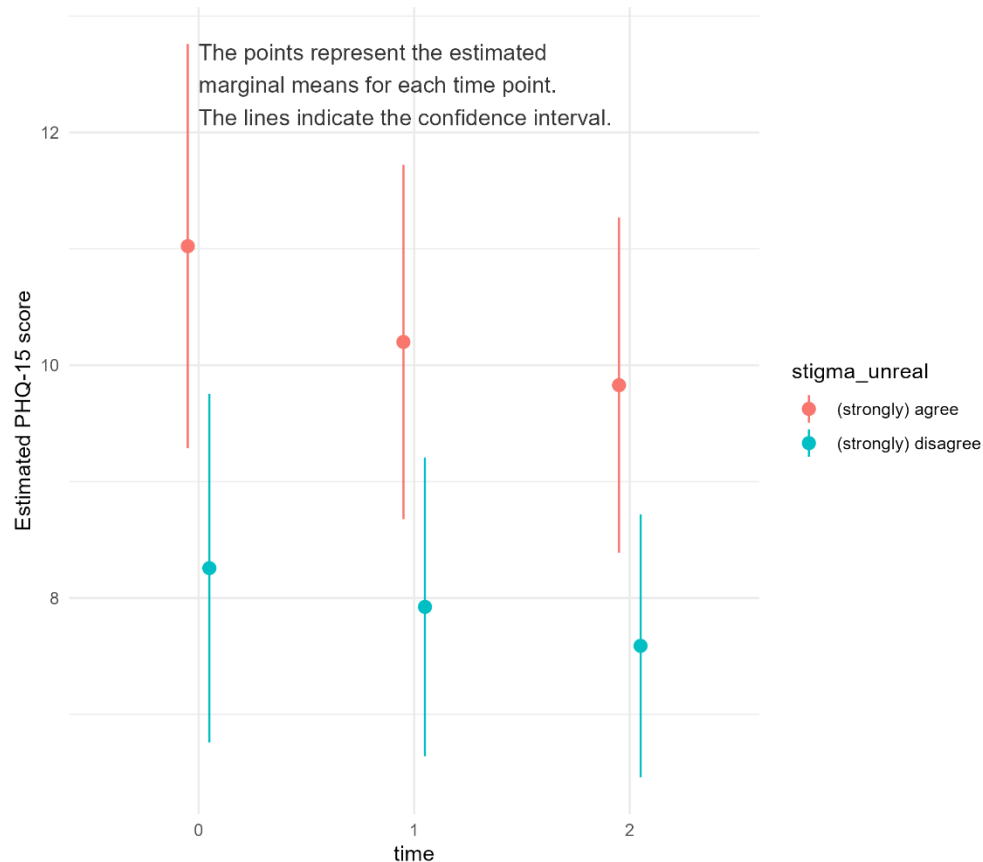
How does symptom severity evolve over time across different stigma groups?



Estimated Marginal Means

Run: `estimate_means(m1, by = c('time', 'stigma_unreal'))`

This command calculates the estimated marginal means for the interaction between time and stigma group.



```
1 emm <- estimate_means(
2   m1,
3   by = c("time", "stigma_unreal"),
4   estimate = "average"
5 )
6 emm
7 #> Average Predictions
8 #>
9 #> time | stigma_unreal | Mean | SE | 95% CI | z
10 #> -----
11 #> 0 | (strongly) agree | 11.02 | 0.89 | [9.29, 12.76] | 12.44
12 #> 1 | (strongly) agree | 10.20 | 0.78 | [8.68, 11.72] | 13.13
13 #> 2 | (strongly) agree | 9.83 | 0.73 | [8.39, 11.27] | 13.38
14 #> 0 | (strongly) disagree | 8.26 | 0.76 | [6.76, 9.75] | 10.81
15 #> 1 | (strongly) disagree | 7.92 | 0.65 | [6.64, 9.21] | 12.10
16 #> 2 | (strongly) disagree | 7.59 | 0.58 | [6.46, 8.72] | 13.16
17 #>
18 #> Variable predicted: phq15
19 #> Predictors modulated: time, stigma_unreal
```

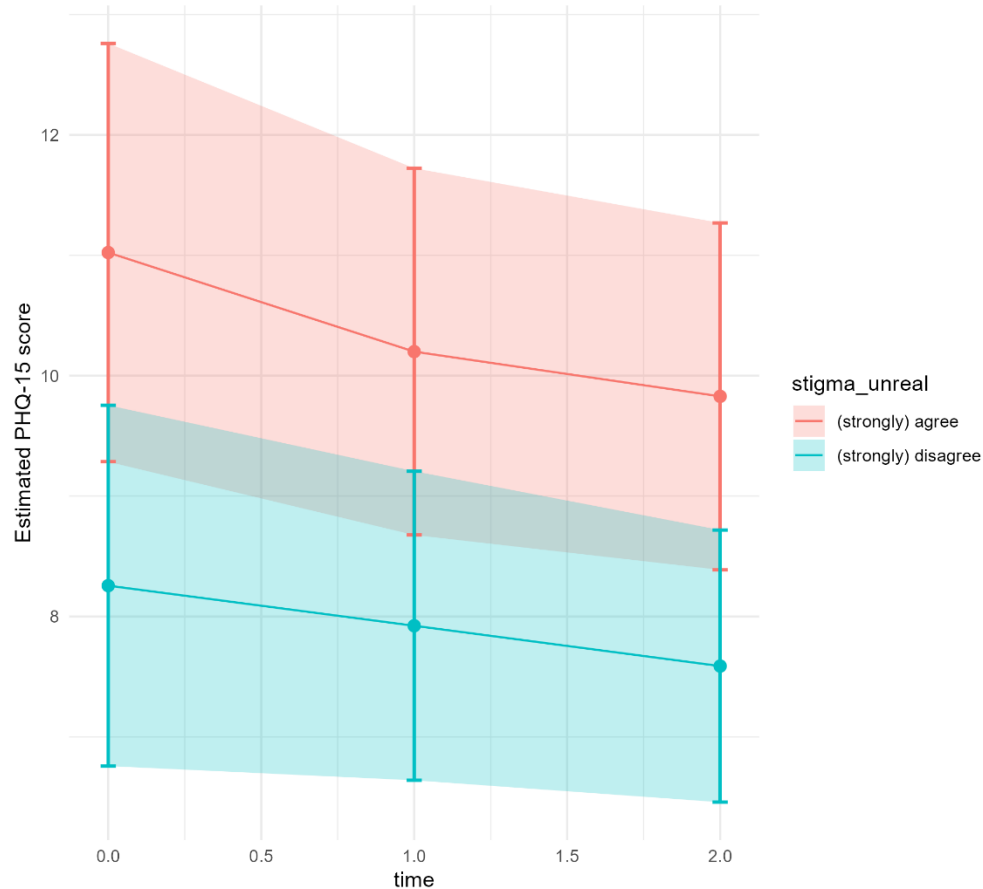
Predicted PHQ-15 trajectories: Overlaying modeled trend

How does symptom severity evolve over time across different stigma groups?



Estimated trend of PHQ-15 scores

The lines show the modeled trend, the points represent the estimated marginal means including 95% confidence intervals for time points 0, 1, and 2.



```

1 emm <- estimate_means(
2   m1,
3   by = c("time", "stigma_unreal"),
4   estimate = "average"
5 )
6 emm
7 #> Average Predictions
8 #>
9 #> time | stigma_unreal | Mean | SE | 95% CI | z
10 #> -----
11 #> 0 | (strongly) agree | 11.02 | 0.89 | [9.29, 12.76] | 12.44
12 #> 1 | (strongly) agree | 10.20 | 0.78 | [8.68, 11.72] | 13.13
13 #> 2 | (strongly) agree | 9.83 | 0.73 | [8.39, 11.27] | 13.38
14 #> 0 | (strongly) disagree | 8.26 | 0.76 | [6.76, 9.75] | 10.81
15 #> 1 | (strongly) disagree | 7.92 | 0.65 | [6.64, 9.21] | 12.10
16 #> 2 | (strongly) disagree | 7.59 | 0.58 | [6.46, 8.72] | 13.16
17 #>
18 #> Variable predicted: phq15
19 #> Predictors modulated: time, stigma_unreal
  
```

Quantifying the change (trend) in PHQ-15 scores: marginal effects

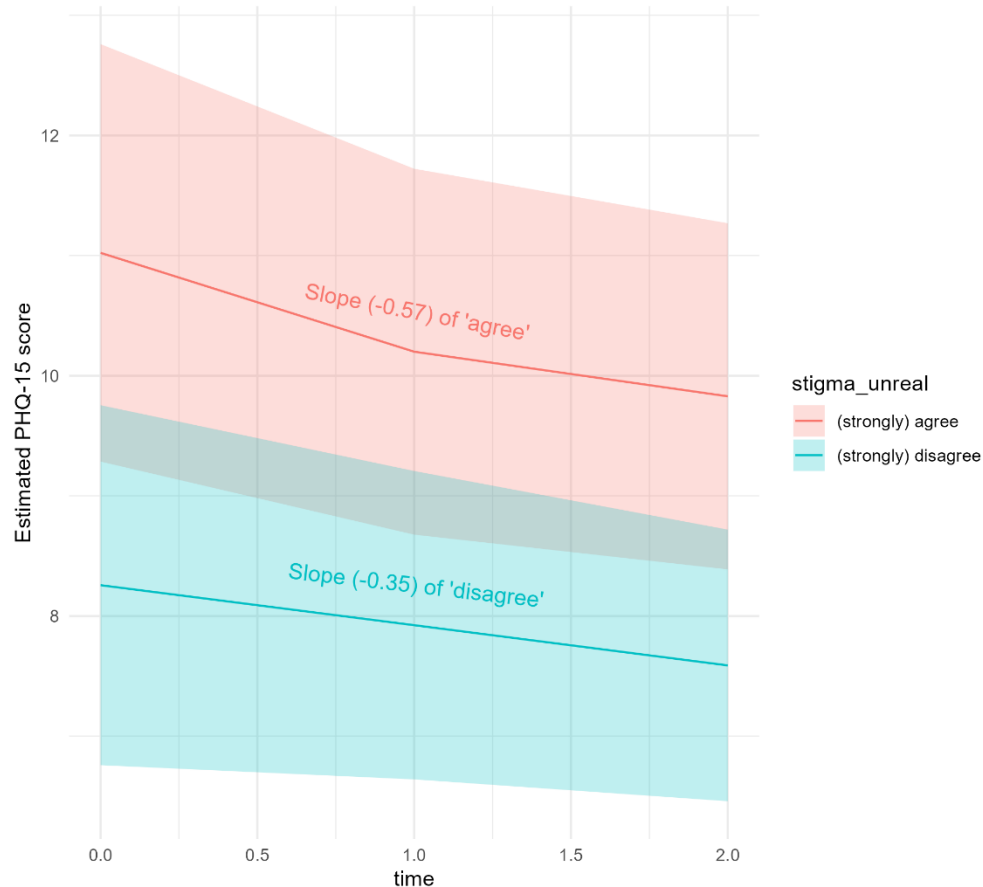
How strong is the time trend in symptom severity within each stigma group?



Estimated trends (slopes) over time

Run: `estimate_slopes(m1, 'time', by = 'stigma_unreal')`

Calculates the average linear slope of the PHQ-15 score per time unit.



```
1 slopes <- estimate_slopes(
2   m1,
3   "time",
4   by = "stigma_unreal",
5   estimate = "average"
6 )
7 slopes
8 #> Estimated Marginal Effects
9 #>
10 #> stigma_unreal      | Slope |   SE |      95% CI |    z |    p
11 #> -----
12 #> (strongly) agree   | -0.57 | 0.24 | [-1.05, -0.09] | -2.33 | 0.020
13 #> (strongly) disagree | -0.35 | 0.17 | [-0.68, -0.01] | -2.01 | 0.044
14 #>
15 #> Marginal effects estimated for time
16 #> Type of slope was dY/dX
```

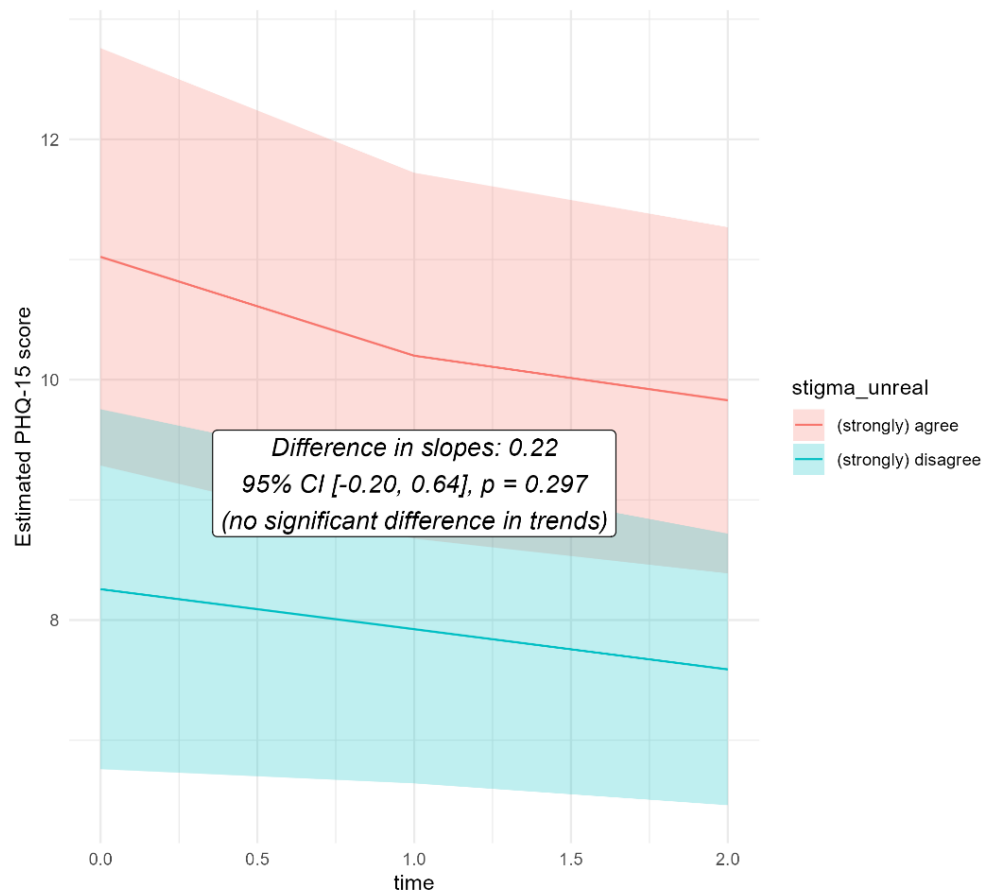


Testing for slope differences

Does the rate of change depend on stigma?

Comparison of Trends

Run: `estimate_contrasts(m1, 'time', by = 'stigma_unreal', integer_as_continuous = TRUE)`
Tests whether the difference between the two slopes is significantly different from zero.



By default, when numeric predictors contain only few unique values (like in this example *time* with three different values), they are treated as “categorical” when contrasts are calculated. `integer_as_continuous` makes sure that we are always calculating contrasts of slopes.

```
1 contrasts <- estimate_contrasts(
2   m1,
3   "time",
4   by = "stigma_unreal",
5   integer_as_continuous = TRUE,
6   estimate = "average"
7 )
8 contrasts
9 #> Averaged Contrasts Analysis
10 #>
11 #> Level1 | Level2 | Difference | SE | 95% CI | z | p
12 #> -----
13 #> strongly disagree | strongly agree | 0.22 | 0.21 | [-0.20, 0.64] | 1.04 | 0.297
14 #>
15 #> Variable predicted: phq15
16 #> Predictors contrasted: time
17 #> p-values are uncorrected.
```

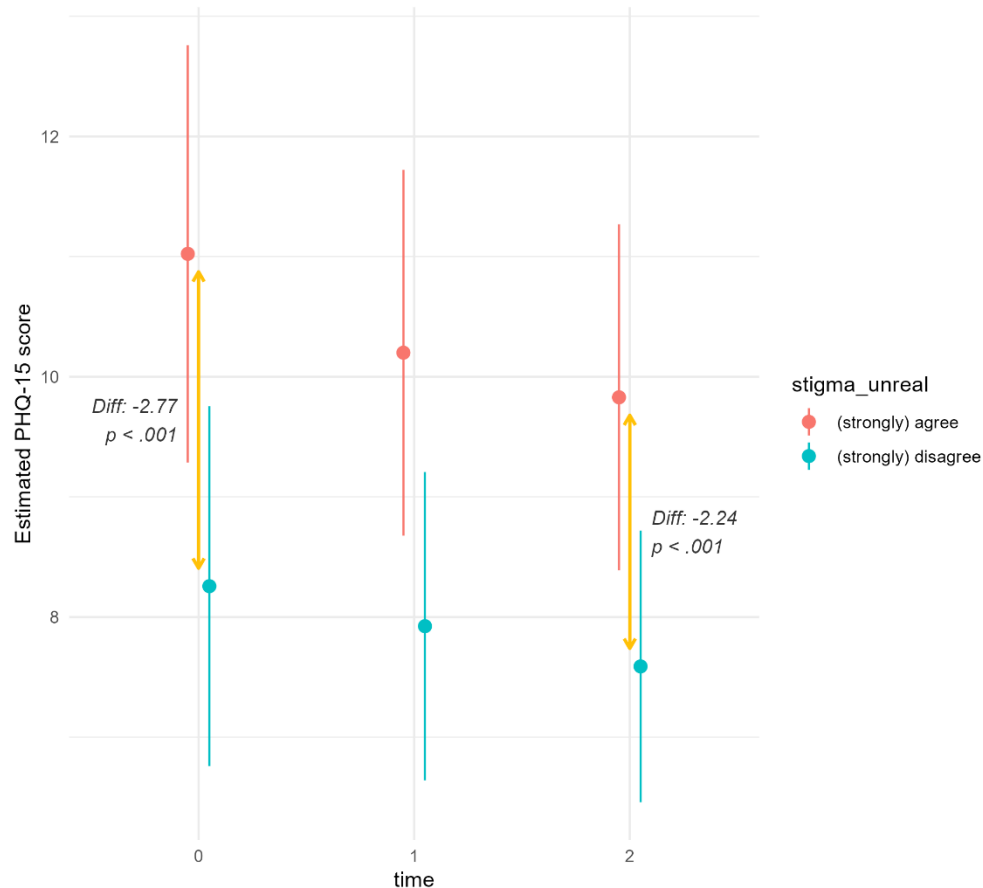
Differences between stigma groups at specific times

Is there a significant gap between the groups at specific measurement points?



Group differences at the beginning and end

Run: `estimate_contrasts(m1, 'stigma_unreal', by = 'time=c(0, 2)')`
Tests the difference between the groups at time points 0 and 2.



```
1 contrasts <- estimate_contrasts(
2   m1,
3   "stigma_unreal",
4   by = "time=c(0,2)",
5   estimate = "average"
6 )
7 contrasts
8 #> Averaged Contrasts Analysis
9 #>
10 #> Level1          | Level2          | time | Difference | SE | 95% CI | z | p
11 #> -----
12 #> strongly disagree | strongly agree | 0 | -2.77 | 0.53 | [-3.81, -1.72] | -5.19 | < .001
13 #> strongly disagree | strongly agree | 2 | -2.24 | 0.55 | [-3.32, -1.16] | -4.06 | < .001
14 #>
15 #> Variable predicted: phq15
16 #> Predictors contrasted: stigma_unreal
17 #> p-values are uncorrected.
```

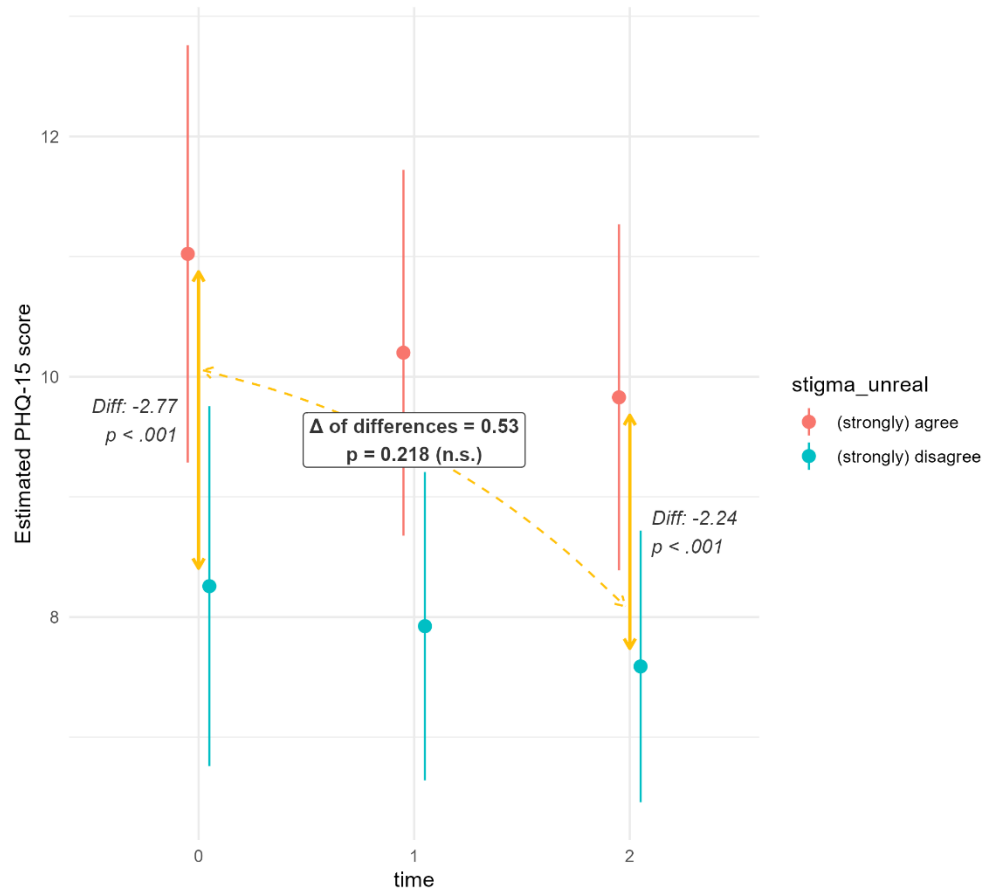


Testing the interaction

Does the group gap change over time?

Comparison of group differences over time

Run: `estimate_contrasts(m1, c('time', 'stigma_unreal'), comparison = '(b1 - b4) = (b3 - b6)')`
 Tests whether the group difference at time=0 differs significantly from the difference at time=2.



```
1 contrasts <- estimate_contrasts(
2   m1,
3   c("time", "stigma_unreal"),
4   comparison = "(b1 - b4) = (b3 - b6)",
5   estimate = "average"
6 )
7 contrasts
8 #> Averaged Contrasts Analysis
9 #>
10 #> Parameter | Difference | SE | 95% CI | z | p
11 #> -----
12 #> b1-b4=b3-b6 | 0.53 | 0.43 | [-0.31, 1.36] | 1.23 | 0.218
13 #>
14 #> Variable predicted: phq15
15 #> Predictors contrasted: time, stigma_unreal
16 #> p-values are uncorrected.
17 #> Parameters:
18 #> b1 = time [0], stigma_unreal [(strongly) agree]
19 #> b4 = time [0], stigma_unreal [(strongly) disagree]
20 #> b3 = time [2], stigma_unreal [(strongly) agree]
21 #> b6 = time [2], stigma_unreal [(strongly) disagree]
```

Questions?

